Issue	**1**
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Usman & Sons

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

Plot No. A-95, S.I.T.E. Super Highway, Phase 1
75700 Karachi
Sindh Pakistan

Latitude: [25.004252], longitude: [67.09787]
Date of inspection: 05-May-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 05-05-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 05-May-2026, Syed Hassan Hussain and Zeeshan Haider on behalf of the Pakistan Accord carried out an initial electrical inspection for Usman & Sons, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Feroz Ahmed (CO), Mr. Amin Usman (GM Technical), Mr. Imran Usman (CEO), Mr. Zuhaib Ali (Compliance representative), Mr. Waqas (Electrical Engineer) and other staff members of Usman & Sons. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were present.

The factory operates in one shift for all departments. Approximately 120 workers operate in the factory with 5% workers being female.

The factory comprises one main building and 4 ancillary buildings owned by Usman & sons. During the inspection, the buildings highlighted in Figure 1.1 were inspected, with the site boundary indicated in red.

- **Main building** is a 5 storey (Ground + mezzanine + 3). The building is used for stitching and storage (3rd, 2nd and 1st floor), offices and storage (mezzanine floor) and stitching and storage (ground floor). The construction date of building is 2001.
- **Ancillaries:**
- **Masjid** is a single storey (Ground only). The building is used for prayers (ground floor). The construction date of building is 2001.
- **Gate office** is a single storey (Ground only). The building is used for security office, fire pump room and generator room (ground floor). The construction date of building is 2001.
- **B1** is a 2 storey (Ground + 1). The building is currently non-operational, and factory has decided to demolish the structure.
- **B2** is a single storey (Ground only). The building is currently non-operational, and factory has decided to demolish the structure.

A summary of the main concerns in relation of electrical safety systems identified during the inspection (non-exhaustive) that should be actioned are as follows:

- **Fire Alarm System:** The facility does not have a proper compliant fire alarm system. This creates significant fire safety risks, including delayed fire detection, absence of automatic initiation from smoke or heat, no early warning for occupants, and increased potential for fire spread before evacuation measures can be taken.

- **Emergency Lights:** Insufficient emergency lights and directional signage were found throughout the facility. These issues pose a serious risk to safe evacuation of workers during an emergency.

- **Earthing System:** The earthing system was non-compliant with the standard due to inaccessible earth pits and undersized earth cables. In addition, bonding of cable trays, metal structures, doors, and metallic furniture/fittings were absent. These pose significant safety risks, such as electric shock, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the action identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

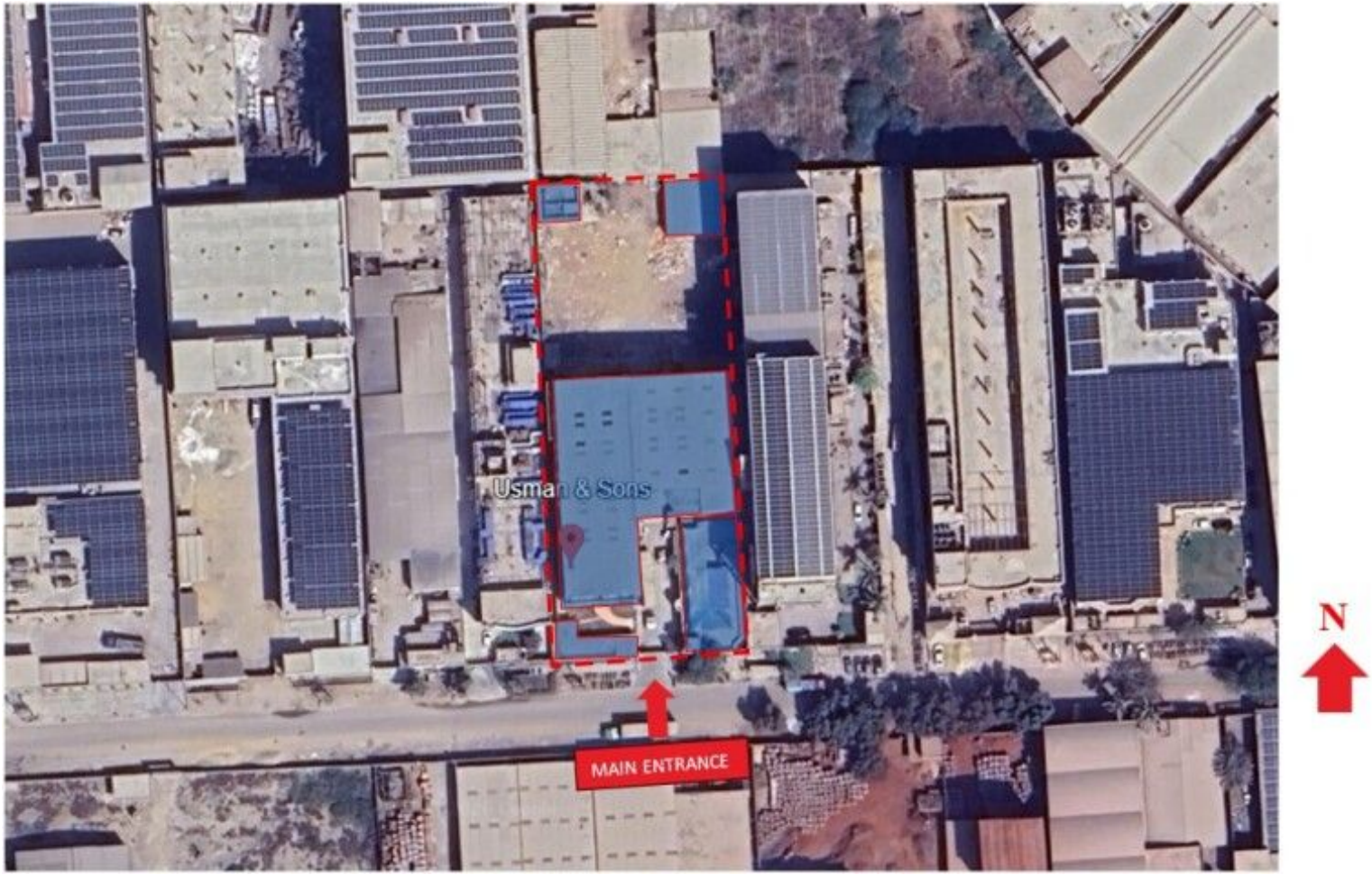


Figure 1.1. Site plan (Satellite view) - Red dashed line indicates site boundary, highlighted (blue) area indicated area inspected.

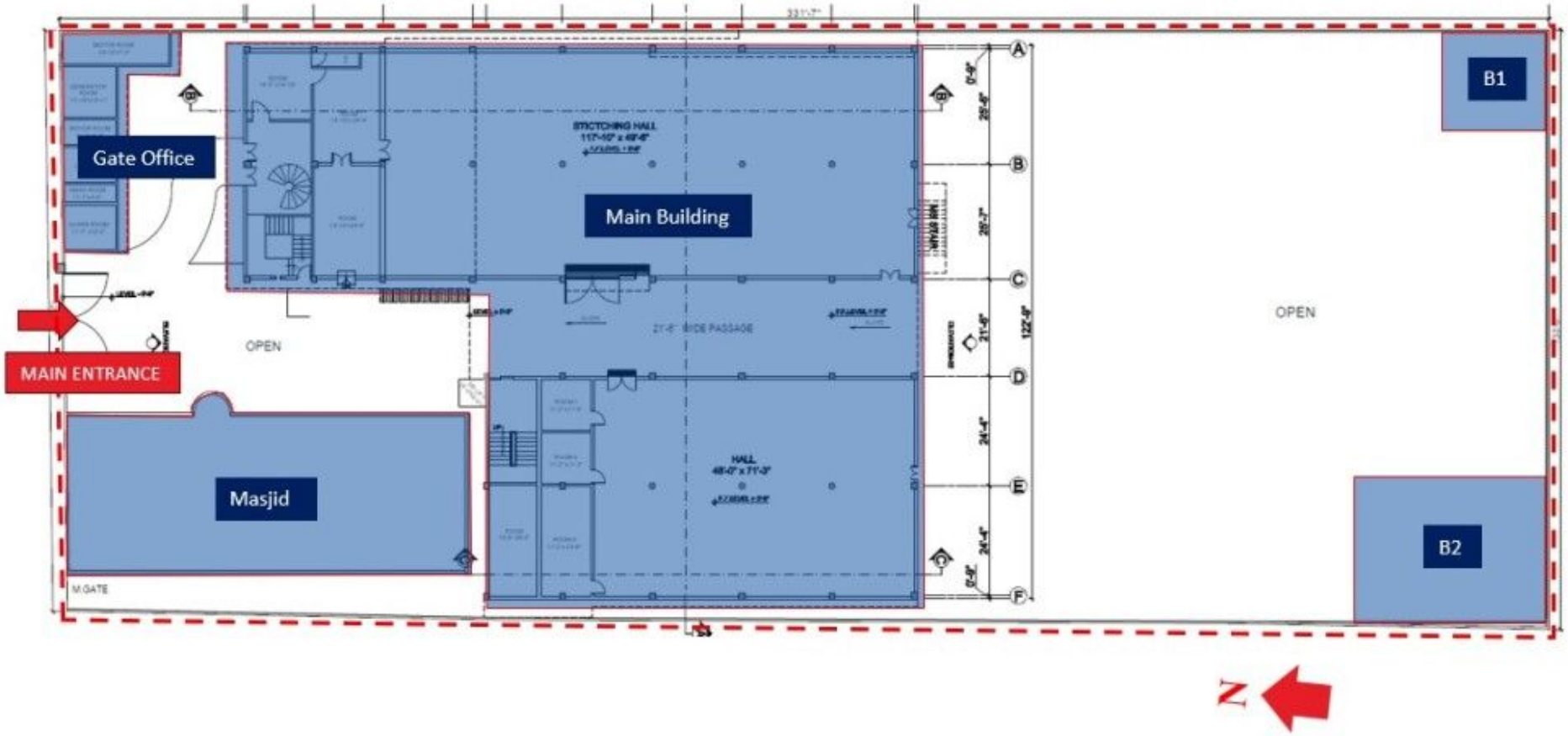


Figure 1.2. Site layout. Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected.

2.1 Building Use

The factory comprises one building, and four ancillary buildings the building uses per floor are as follows:

Main Building-01:
L00 - Ground floor: Stitching and storage.
M01 - Mezzanine: Offices and storage.
L01 - First floor: Stitching.
L02 - Second floor: Stitching.
L03 - Third floor: Stitching.

Ancillaries:
Masjid: Ground floor
Security Office: Ground floor
B1: Not occupied
B2: Not occupied

2.2 Utility Source

The factory receives a low-voltage electrical supply from Karachi Electric (65 kW sanctioned load) via 150kVA PMT placed outside the factory, which feds their main LT panel, and then low-voltage power is distributed to the respective floors distribution boards via cables.



Figure 1.3. 150kVA transformer for factory primary power source

2.3 Backup source

The factory has one diesel generator having a rating 100kW that serves as a backup power source. Additionally, the emergency lights and fire alarm system are equipped with built-in battery bank to provide backup during power outages.



Figure 1.4. 100kW diesel generator for backup power source

3. Documentation

This table shows the details of the documentation provided by the factory:

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	Yes		The submitted drawing was incomplete and missing necessary details.
2	Single Line Diagrams	Yes		The submitted drawing was incomplete and missing necessary details.
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of Usman & Sons, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Usman & Sons. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202605-066509

Timeline	Remediation Plan
Medium Term CAP-0079372	Factories should begin remediation of CAP items such as cleaning, cable organisation, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no electrical room layout drawings available for the factory.

There are no visible indications of LT panels equipment locations or pathways for maintenance and emergency access.
The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and an increased likelihood of equipment failure. F-202605-066502

Timeline	Remediation Plan
Medium Term CAP-0079376	Develop detailed electrical room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079377	Regularly update and review the electrical room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were available but not updated and missing necessary details such as (Outgoing circuits cable sizes are not mentioned in drawings, bus bar rating is not mentioned in drawing).

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202605-066495

Timeline	Remediation Plan
Medium Term CAP-0079380	Update comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079381	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory has available electrical layout drawings, including power layouts, lighting layouts and containment layouts. However, the drawings did not contain necessary details, such as circuit details with sizing in the layouts, types and locations of outlets, standard symbols, and the locations of power distribution boards.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202605-066505

Timeline	Remediation Plan
Medium Term CAP-0079378	Update comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079379	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.



Figure 2.1 - Electrical layout drawings

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.
During the survey, the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

i. It was found that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes such as:
a) Ground and first floor distribution board 125A circuit breaker with 16sq.mm cable.

Circuit breakers that are not properly sized may fail to trip during overload conditions, allowing excessive current to flow and causing overheating, which leads to a fire risk.

ii. It was also found at multiple locations that single-phase outgoing was used from three pole Breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system. Observed at
a) Ground floor LT room panels.

iii. It was observed that the circuit breakers were installed in the facility without proper fixing supports and protective enclosures. This inadequate installation could lead to loose connections, mechanical damage, accidental contact with live parts, electrical faults, and an increased risk of fire or electric shock incidents.
a) Observed at mezzanine floor offices area.F-202605-066491

Timeline	Remediation Plan
Medium Term CAP-0079392	i. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.

Medium Term CAP-0079393	ii. Remove the single-phase from the three-pole breaker and use a single-pole breaker for single-phase connection out as per standard cable current carrying capacity.
Medium Term CAP-0079394	iii. Fix all circuit breakers that are loosely installed throughout the facility with metallic enclosures to ensure safety and compliance with standards BS 7671 Section 416.



Figure 2.2 - Single phase connection taken out from three pole mccb breaker found at ground floor LT room panel



Figure 2.3 - Inadequate breaker rating was found at ground floor LT panels 125A circuit breaker with 16Sq.mm cable



Figure 2.4 - Inadequate breaker rating was found at third floor LT panels 125A circuit breaker with 16Sq.mm cable



Figure 2.5 - Circuit breaker was found without fixing supports and protective enclosure at mezzanine floor

7.2 Hot Spots

This relates to the entire facility.
During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey detected several hot spots in multiple distribution boards, observed at:
a) Ground floor LT panel (Temperature 58.7 °C)
b) First floor office area distribution board (Temperature 56.7 °C).

Hot spots in electrical distribution systems can be caused by several factors, including:
> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation
All temperature related assessments are based on Table 100.18 of ANSI Standard 2019.

It was also found that the thermographic test report was not available at the site.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202605-066493

Timeline	Remediation Plan

Medium Term CAP-0079395	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0079396	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.



Figure 2.6 - Hot spot was found at ground floor LT room panel



Figure 2.8 - Hot spot was found at first floor office distribution board pvc joint



Figure 2.7 - Hot spot was found at ground floor LT room panel



Figure 2.9 - Hot spot was found at first floor office distribution board pvc joint

7.3 Earthing system

This section relates to the entire facility.
During the inspection of the facility's electrical panels, distribution boards, and machinery, the following observations with the earthing system were identified.

i. It was also observed that the factory does not have an earthing layout drawing available.

ii. It was found that the earth circuits distribution was missing in multiple distribution boards.
Observed at,
a) Ground floor LT panels.
b) Ground, first and second floor distribution boards.

iii. It was also found that the equipotential bonding for the distribution board front doors, cable trays, machines body, metallic cage of generator and panel covers were missing.
Observed at,
a) Ground floor LT panels.
b) Ground, first and second floor distribution boards.

iv. The earth pits were inaccessible and unsuitably located at ground floor utility area of the facility.

v. No earth pit testing report was available.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202605-066494

Timeline	Remediation Plan
Medium Term CAP-0079406	i. Develop and submit the earthing system design to the Pakistan Accord for review.

Longer Term CAP-0079407	ii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Longer Term CAP-0079408	iii. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Longer Term CAP-0079411	iv. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Longer Term CAP-0079679	v. Submit the earth testing report to Pakistan Accord for review.
Longer Term CAP-0079412	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.





Figure 2.10 - Earth distribution and bonding connection were found missing inside ground floor distribution board



Figure 2.12 - Earth pit was found inaccessible at ground floor utility area



Figure 2.14 - Earth wire was found missing with power socket at first floor kitchen area

Figure 2.11 - Earth distribution and bonding connection were found missing inside second floor distribution board



Figure 2.13 - Earth distribution and bonding connection were found missing inside first floor office area distribution board



Figure 2.15 - Two pin power plug used for all of the stitching machines in facility without earth wire



Figure 2.16 - Earth distribution and bonding connection were found missing inside first floor main distribution board

7.4 Panel/cabling/busbar

This section relates to the entire facility.
The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. It was found that several distribution boards and cable terminals exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements.
Observed at,
a) Second floor lift control panel.

ii. It was also found that there was loose wiring and insufficient working space inside the distribution boards for operation and maintenance. Loose wiring and insufficient space, lead to excessive heat generation which can cause the insulation to melt and potentially lead to fire.
Observed at ground floor office area distribution board.F-202605-066492

Timeline	Remediation Plan
Medium Term CAP-0079399	i. Replace or refurbish corroded electrical distribution boards to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0079400	ii. Install wire ducts to organize and route wires neatly, remove loose cables, and ensure that no cables are attached to the bus bars. This may mean the addition of new distribution boards with proper accessibility for cables incoming and outgoing, bus-bar alignment.
Longer Term CAP-0079402	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.17 - Corroded circuit breaker terminals were found inside second floor lift distribution board



Figure 2.18 - Loose wiring was found inside office area distribution board of ground floor

7.5 Loop Connection

This section relates to the entire facility.

During the inspection loop connections were identified within the electrical distribution boards.
In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.
Observed at ground, first and second floor distribution boards.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202605-066498

Timeline	Remediation Plan
Medium Term CAP-0079388	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.19 - Loop connection was found in first floor distribution board



Figure 2.20 - Loop connection was found in ground floor distribution board

7.6 Multiple Connections

This section relates to the entire facility.

During our inspection, multi looping was observed in a number of electrical panels across the facility. In these cases, multiple circuits were being protected by a single circuit breaker.
Observed at ground and second floor distribution boards.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202605-066497

Timeline	Remediation Plan
Medium Term CAP-0079389	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.21 - Multiple connections were found at ground floor production area distribution board



Figure 2.22 - Multiple connections were found at ground floor production area distribution board



Figure 2.23 - Multiple connections were found at second floor lift control distribution board

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

It was found that many distribution boards have openings, which may cause potential hazards due to environmental factors.
Observed at,
a) Ground floor main LT panels.
b) Ground, first and second floor distribution boards.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202605-066496

Timeline	Remediation Plan
Medium Term CAP-0079390	Examine the entire facility panels to identify all openings, holes, and gaps that need to be sealed. Seal the openings, holes of panels, cable trays and trenches with metallic enclosures, gaskets, or fillers, use cable glands for cable entry. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0079391	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.24 - Unsealed opening was found inside first floor distribution board



Figure 2.25 - Unsealed opening was found inside ground floor main LT panels

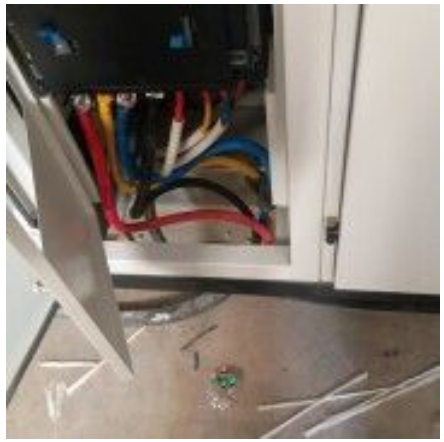


Figure 2.26 - Unsealed opening was found inside ground floor main LT panels



Figure 2.27 - Unsealed opening was found inside third floor distribution board



Figure 2.28 - Unsealed opening was found inside first floor distribution board

7.8 Adequate accessibility

This section relates to the entire facility.
During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At multiple locations, it was found that the electrical distribution boards do not possess the required space for inspection or easy access to the panels and their components.
Observed at ground floor generator room distribution board.

Limited accessibility to distribution boards and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202605-066500

Timeline	Remediation Plan
Medium Term CAP-0079385	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.29 - Inadequate accessibility was found for main distribution board in generator room

7.9 Lint & dust inside panel board

This section relates to the entire facility.
During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in few of the cable trays and distribution boards.
Observed at,
a) Ground and second floor area distribution boards.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.F-202605-066501

Timeline	Remediation Plan
Medium Term CAP-0079382	All electrical panels, cable trays and distribution boards shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0079383	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and distribution boards. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.30 - Lint and dust was found inside ground floor distribution board



Figure 2.31 - Lint and dust was found inside second floor lift control distribution board

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was found that no lightning protection system was installed, and there was no risk assessment provided to confirm whether it was not required.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202605-066499

Timeline	Remediation Plan
Medium Term CAP-0079386	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment based on BS EN 62305. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0079387	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire building.
During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

It was observed that an conventional and non listed fire alarm system has been installed within the facility.

Furthermore, several deficiencies were identified during the inspection. These include inadequate detection coverage in the throughout the entire building. In addition, a number of detection devices were found to be obstructed by structural beams and other physical barriers, potentially affecting their performance.
Moreover, non-fire rated cable was used throughout the fire alarm system. F-202605-066503

Timeline	Remediation Plan
Medium Term CAP-0079416	(i) Submit design documentation for new replacement system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets and product certifications of all listed equipment to show full compliance of the Fire Alarm and Detection System (FADS) with NFPA 72.
Medium Term CAP-0079415	(ii) Submit documentation to Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0079417	(iii) Submit documentation to Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on site for installation.
Longer Term CAP-0079418	(iv) Once installation of the FADS system is complete, factory to submit full "as-built" documentation of the system to Pakistan Accord for review.
Longer Term CAP-0079419	(v) Factory to advise Pakistan Accord when FADS installation fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.

Longer Term CAP-0079420	(vi) Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.32 - Insufficient fire detector coverage was found at second floor production area





Figure 2.33 - Conventional and non listed fire alarm control panel



Figure 2.35 - Fire detector was found missing in ground floor office

Figure 2.34 - Non fire rated cable used throughout the building fire alarm system



Figure 2.36 - Fire detector was found missing in first floor office area

room



Figure 2.37 - Independent type detectors was found at ground floor LT room

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations such as:
a) Ground floor production and storage area emergency lights.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202605-066506

Timeline	Remediation Plan
Medium Term CAP-0079427	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0079428	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.38 - Emergency light was found non-functional at ground floor storage area



Figure 2.39 - Emergency light was found non-functional at ground floor storage area

9.2 Provided but insufficient

The section refers to the entire facility.
During the inspection the emergency lighting arrangement throughout the facility was assessed.
The following observations were made:

i. It was also observed that the emergency lighting layout drawing was not available at the site.

ii. During the inspection, it was found that the emergency lights and signage provided in the facility were insufficient to cover the entire facility, leading to dark or poorly lit areas during power outages. This inadequate coverage may cause incidents during emergencies.
Observed at,
a) Main building all floors production area, offices, warehouses, stores, electric rooms and stairs.

iii. External emergency lighting was not provided for the facility, preventing people from reaching a place of safety outside the facility during nighttime evacuations. Observed at main building outdoor area.

iv. The backup power supply to the emergency luminaires is insufficient in different areas such as main building ground floor area emergency lights.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202605-066507

Timeline	Remediation Plan
Medium Term CAP-0079433	i. Submit the emergency lighting system design to the Pakistan Accord for review.
Medium Term CAP-0079434	ii. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Longer Term CAP-0079435	iii. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s).

Longer Term CAP-0079436	iv. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term CAP-0079437	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.40 - Emergency light were found missing at main building stairs



Figure 2.41 - Emergency light were found missing at ground floor store room



Figure 2.42 - Emergency light were found missing at first floor office areas



Figure 2.43 - Emergency light were found missing at building outdoor areas



Figure 2.44 - Insufficient emergency lights were found at second floor production area



Figure 2.45 - Emergency light were found missing at first floor account office areas



Figure 2.46 - Insufficient emergency lights were found at third floor production area



Figure 2.47 - Emergency light were found missing at main building outdoor areas

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system according to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202605-066504

Timeline	Remediation Plan
Medium Term CAP-0079425	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately when general lighting power fails, as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0079426	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

i. It was found that the battery terminals were exposed, Corroded and not properly insulated, bare terminals increase the risk of electrical shock or short circuits. potentially igniting fires or explosions. Proper monthly and annual maintenance of the battery system is crucial for fire safety. Observed at the ground floor generator room.

ii. The placement of generator batteries did not comply with safety regulations, potentially hindering safe access and proper ventilation.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202605-066508

Timeline	Remediation Plan
Medium Term CAP-0079440	i. Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0079441	i. Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.
Medium Term CAP-0079442	ii. Reposition batteries that are improperly placed to ensure they comply with safety regulations, allowing for safe access and adequate ventilation. Mark and clear pathways around battery installations to ensure easy and safe access for maintenance and emergencies.
Longer Term CAP-0079443	ii. Conduct regular audits of battery placements to ensure ongoing compliance with safety standards and regulations.



Figure 2.48 - Generator battery terminals were found corroded and non insulated



Figure 2.49 - Inappropriate placement of ground floor generator batteries

11. Cabling & Wiring

11.1 Cable support and protection

This section relates to the entire facility.
During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

It was observed that cables were installed without proper containment systems such as cable trays, conduits, or ducting, and were laid directly on the floor. The absence of adequate cable containment and support can cause sagging or mechanical stress on the cables, potentially leading to damage or failure.
Observed at,
a) Ground floor generator and LT room.
b) Ground floor corridor area main cable running over floor.
c) Second floor production area cutting machines cables.

Cables on the floor are susceptible to mechanical damage from foot traffic, material handling equipment, and other operations, which could result in electrical short circuits, potential electric shock, or even fire. Additionally, the exposed cables can cause workers to trip and fall, leading to possible injuries.F-202605-066487

Timeline	Remediation Plan
Medium Term CAP-0079448	Secure cables with appropriate containment and supports to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Longer Term CAP-0079449	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.50 - Cable was found without containment and placed directly over floor at second floor production area



Figure 2.51 - Cables were found without containment at ground floor LT room



Figure 2.52 - Cables were found without containment at ground floor LT room



Figure 2.53 - Cables were found without containment at ground floor corridor area



Figure 2.54 - Cables were found without containment at first floor storage room



Figure 2.55 - Cable was found without containment at ground floor generator room



Figure 2.56 - Cable was found without containment at ground floor generator room

11.2 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Several trunking systems were found to be filled with fluff, which is not standard practice and poses risks related to cable overheating, moisture ingress, and potential electrical faults. Observed at the first floor production area cable trays.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris.F-202605-066489

Timeline	Remediation Plan
Medium Term CAP-0079452	Remove all fluff from the cable trunking and clean the system to ensure it remains dry, clear, and compliant with safety standards. Install proper covers or sealing measures to prevent sand or debris from entering in the future.
Longer Term CAP-0079453	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.57 - fluff was found inside first floor production area cable trays

11.3 Splicing of cables

This section relates to the entire facility.
During the inspection, the cable installation was visually surveyed throughout the facility. The following observations were noted:

i. It was found that cable terminations were made without using cable lugs, leading to broken strands and loose connections.
Observed at,
a) Ground, first and second floor distribution boards..

ii. PVC tape joints were found at multiple locations. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical short circuits and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks.
Observed at,
a) Ground floor LT room cable.
b) Ground, first and second distribution board and machines cable.

iii. It was also found that the non-compliant PVC connectors were used for circuit connection. Non compliant PVC connectors can hold dust and fluff, which can ignite fire in terms of short circuits.
Observed at,
a) First floor distribution board area.

iv. At many locations exposed electrical connections were also found. These exposed connections increase the risk of electric shock, short circuits, and potential fire hazards. Observed at,
a) Ground floor production area.

v. It was also found that at ground floor store area wires were used for power connection without a power plug. This practice is highly dangerous as it can lead to loose connections, electrical shocks, and short circuits. Direct insertion bypasses proper safety mechanisms, significantly increasing the risk of fire and electrical hazards, thereby endangering life safety.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety.F-202605-066490

Timeline	Remediation Plan
Medium Term CAP-0079459	i. Install cable lugs on all improperly terminated cables within the distribution boards to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0079460	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Medium Term CAP-0079461	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with BS7671 18th Edition + A2 (2022).
Longer Term CAP-0079462	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Medium Term CAP-0079463	iii. Ensure complete replacement of all PVC connectors with compliant terminations as per BS 7671 and train staff on approved practices.
Longer Term CAP-0079464	iii. The factory needs to develop and implement a standard operating procedure (SOP) for the proper use of terminal blocks and to prevent temporary or unsafe electrical joints.
Medium Term CAP-0079465	iv. Remove or insulate the exposed connections with heat shrink tubing or quality insulation tape to bare wire connections. Ensure that the insulation material is durable and rated for the voltage level of the circuit.
Medium Term CAP-0079466	v. Replace the directly inserted power wires with proper power plugs to ensure safe and secure connections.
Longer Term CAP-0079467	v. Develop and enforce a facility-wide electrical safety protocol that mandates the use of power plugs for all equipment. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards.



Figure 2.58 - Machine wires were directly used for power connection without a power plug found in ground floor store room



Figure 2.60 - PVC tape joints were found cable outgoing to floor distribution board from main LT panel



Figure 2.59 - Non compliant PVC connector was found at first floor main distribution board room



Figure 2.61 - PVC tape joint was found at third floor production area machine



Figure 2.62 - Exposed connection was found in production area ground floor



Figure 2.63 - PVC tape joint was found at first floor production area main distribution board



Figure 2.64 - Cables were found terminated without cable lugs at second floor distribution board



Figure 2.65 - Exposed and PVC tape joint were found in second floor lift distribution board



Figure 2.66 - Cables were terminated without cable lugs at ground floor distribution board

11.4 Colour coding

This section relates to entire facility.

During the inspection of multiple distribution boards, deviations from standard cable colour coding practices were identified.
Observed at,
a) Ground floor office, first floor lift control panel and second floor lift distribution board.

Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure the accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202605-066486

Timeline	Remediation Plan
Medium Term CAP-0079450	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labeling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0079451	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices.



Figure 2.67 - Colour coding violation was found at ground floor office distribution board



Figure 2.68 - Colour coding violation was found at first floor main distribution board



Figure 2.69 - Colour coding violation was found at second floor lift control distribution board

11.5 Fixtures

This section relates to the entire facility.
During the inspection, the electrical fixtures throughout the facility were visually surveyed. The following observations were noted.

i. Wooden switch board were found at many locations. Wood is flammable, and if the electric circuit inside the wooden switchboard generates excessive heat due to overloads or faults, it can ignite the wooden box. This can lead to a fire hazard.
Observed at,
a) First floor production area.

ii. Lighting fixtures were observed to be improperly installed at multiple locations, where they are hung or supported directly by electrical cables instead of adequate mechanical supports. Unsupported or loosely hung fixtures may fall, causing injury to personnel below.
a) Second floor production area light bulb supported by cable.

iii. Broken and faulty fixtures were found in many locations. This poses a significant risk of equipment damage or electrical incidents due to the instability of the fixture. Loose connections can lead to electrical arcing, which increases the risk of fire and can cause serious injury or equipment failure.
Observed at
a) Ground and first floor office area.F-202605-066488

Timeline	Remediation Plan
Medium Term CAP-0079454	i. Replace the wooden switchboard with a non-combustible enclosure designed for electrical installations. Additionally, replace the wooden ply with metallic brackets or stands.
Longer Term CAP-0079455	i. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.
Medium Term CAP-0079456	ii. Install all lighting fixtures with proper mechanical supports such as brackets, chains, or conduits independent of electrical cables.
Medium Term CAP-0079457	iii. Replace and secure the faulty and broken fixture with appropriate fixed supports to prevent movement and ensure stable connections.



Figure 2.70 - Inadequate fixing support were found for light bulb at second floor production area



Figure 2.71 - Broken power receptacle was found at ground floor office room



Figure 2.72 - Wooden switchboard was found at first floor production area



Figure 2.73 - Broken power receptacle was found at first floor office room

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.